

Worksheet 2 — What Does $|\mathbf{R}|$ Actually Measure?

STAT 481 — Multivariate Analysis • Lecture 2 • Thu, Aug 27, 2026

1. Objective of this lecture

By the end of the hour you should be able to:

1. rebuild any entry of the **sample covariance matrix** $\mathbf{S}$ and the **sample correlation matrix** $\mathbf{R}$ from its definition, and certify the whole matrix against `cov()` and `cor()`;
2. predict — exactly, before running it — how $\mathbf{S}$, $\mathbf{R}$, $\text{tr}(\mathbf{S})$ and $|\mathbf{S}|$ respond to a change of units;
3. say what $|\mathbf{R}|$ measures, manufacture a data set with $|\mathbf{R}| = 0$, and locate the redundancy in the real data.

2. The data, and how to get it in R

Data set: `worldhealth2022.csv`, in the course project — no package needed, `read.csv()` is base R.
Size: 176 countries $\times$ 8 indicators (World Bank, 2022). Stage 6 also uses `corrplot` (on CRAN). Open `STAT481.Rproj` first, so the relative path resolves.

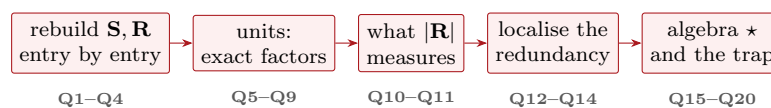
```
wh <- read.csv("data/worldhealth2022.csv", row.names = 1)
W <- wh[, 4:11] # the 8 numeric indicators; cols 1-3 are labels
n <- nrow(W); p <- ncol(W)
c(n = n, p = p)
```

<code>life_exp</code>	life expectancy at birth, years	<code>urban_pct</code>	urban population, %
<code>infant_mort</code>	infant mortality, per 1000 births	<code>gdp_pc</code>	GDP per capita, US\$
<code>fertility</code>	births per woman	<code>health_exp_pct</code>	health spending, % of GDP
<code>pop65</code>	population aged 65+, %	<code>internet_pct</code>	Internet users, %

3. What we want to know today

The question: the notes assert that $|\mathbf{R}| = 0.0012$ — the determinant of the *sample correlation matrix* — means the eight indicators are *redundant*, that fewer variables would carry nearly all the information. *What does $|\mathbf{R}|$ actually measure — and which variables own the redundancy?*

Today: what $|\mathbf{R}|$ does at $p = 2$ (by hand), what it does when a column carries nothing new (we build one), and how much of the 0.0012 one variable owns.



4. The questions

Stage 1 — rebuild the machine, entry by entry

$\mathbf{S}$ is the **sample covariance matrix**, $\mathbf{R}$ the **sample correlation matrix** — both $p \times p$ and symmetric, both built from the $n \times p$ table one *pair of columns* at a time (Definition 2 of today's notes):

$$s_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k), \quad r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}s_{kk}}}, \quad \mathbf{R} = \mathbf{D}^{-1}\mathbf{S}\mathbf{D}^{-1},$$

where $\mathbf{D} = \text{diag}(\sqrt{s_{11}}, \dots, \sqrt{s_{pp}})$, so the third form only says *divide entry (j, k) by $\sqrt{s_{jj}}$ and $\sqrt{s_{kk}}$* . Taking $j = k$ gives s_{jj} , the ordinary variance of column j : $\mathbf{S}$ carries the p variances on its diagonal, $\mathbf{R}$ carries ones. In R: `cov()` and `cor()`. *New bracket today:* a matrix with named rows and columns is addressed like the data frame in Lecture 1, with names in both slots — `S["life_exp", "gdp_pc"]` is the single entry $s_{\text{life_exp, gdp_pc}}$.

Q1. Section 5 of the notes prints $s_{\text{infant.mort,gdp.pc}} = -255\,063.9$: state its units — and the units of $r_{\text{infant.mort,gdp.pc}}$. guess

Q2. Without `cov()`: compute $s_{\text{life.exp,infant.mort}}$ from the entry formula, then check it against `cov()`. R code

Q3. Using entries of **S** only, rebuild $r_{\text{life.exp,infant.mort}}$ and check it against `cor()`. R code

Q4. Certify $\mathbf{R} = \mathbf{D}^{-1}\mathbf{S}\mathbf{D}^{-1}$ for the whole 8×8 in one line. R code

Stage 2 — units — predict the exact factor, then run

A change of units is multiplying a whole column by a constant: dollars $\rightarrow$ thousands of dollars is $x \mapsto x/1000$; call the constant c . *New assignment today:* `Y$gdp.pc <- W$gdp.pc/1000` replaces the column `gdp.pc` of **Y**; assigning to a column name that does not exist yet (as `Z$rural.pct` will, in Q11) creates that column at the right edge of the table.

Q5. `gdp.pc` moves from dollars to thousands: the exact factors picked up by $s_{\text{gdp.pc,gdp.pc}}$, $s_{\text{life.exp,gdp.pc}}$ and $r_{\text{life.exp,gdp.pc}}$? guess

Q6. Verify all three factors in one block. R code

Q7. Same change — the exact factor picked up by $|\mathbf{S}|$: derive it on the 2×2 first. guess

Q8. Verify it on the 8×8 . R code

Q9. Which variable owns $\text{tr}(\mathbf{S})$ — and who owns it after the change of units? R code

Stage 3 — what $|\mathbf{R}|$ measures

Determinant, 2×2 : $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ (§3, Step 5 of the notes — the bars mean determinant, not size).

The two anchors, from §7: $|\mathbf{R}| = 1$ exactly when all correlations are 0, and $|\mathbf{R}| = 0$ exactly when one variable is an exact linear combination of the others. In R: `det()`.

Q10. For $p = 2$, write $|\mathbf{R}|$ in terms of r_{12} ; evaluate it at $r_{12} = 0, \pm 1$ — then exactly for the four students of §3. guess

Q11. Append `rural_pct = 100 - urban_pct` as a 9th column: predict $|\mathbf{R}|$ of the 9×9 , then run it. R code

| Stage 4 — localise the redundancy

Q12. Removing `life_exp` multiplies $|\mathbf{R}| = 0.0012$ by some factor: commit to an order of magnitude. guess

My factor: $\times$

Q13. Remove `life_exp` and compute the factor. R code

Q14. Remove `health_exp_pct` instead; compare the two factors — which removal freed the matrix? R code

| Stage 5 — $\mathbf{X}$, $\bar{\mathbf{x}}$, $\mathbf{S}$ and $\mathbf{R}$ as matrix algebra ($\star$ — a Lecture 6 preview, not examinable)

Everything so far was done one entry at a time. The same three objects have one-line matrix forms, and Lecture 6 (Sun, Sep 13) derives them properly. Write $\mathbf{1}$ for the $n \times 1$ column of ones and $\mathbf{H} = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ for the **centering matrix**. Then

$$\bar{\mathbf{x}} = \frac{1}{n}\mathbf{X}^\top\mathbf{1}, \quad \mathbf{S} = \frac{1}{n-1}\mathbf{X}^\top\mathbf{H}\mathbf{X}, \quad \mathbf{R} = \mathbf{D}^{-1}\mathbf{S}\mathbf{D}^{-1}.$$

In R, `%*%` is matrix multiplication (plain `*` multiplies entry by entry and is *not* the same thing), `t()` transposes, `diag(n)` builds the $n \times n$ identity $\mathbf{I}$, and `solve()` inverts. Today we only *run* these; why they work is Lecture 6's subject.

Q15. $\bar{\mathbf{x}}$ is the p column averages. Which of $\mathbf{X}^\top\mathbf{1}$ and $\mathbf{1}^\top\mathbf{X}$ has the right shape, and what does the other give? guess

Q16. Build $\mathbf{X}$ as a numeric matrix, compute $\bar{\mathbf{x}}$ by matrix algebra, and certify it against `colMeans()`. R code

Q17. $\mathbf{H} = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ is 176×176 . What does $\mathbf{H}\mathbf{X}$ do to each column, and why does that make $\mathbf{X}^\top\mathbf{H}\mathbf{X}$ the right thing to divide by $n - 1$? guess

Q18. Compute $\mathbf{S}$ and $\mathbf{R}$ by matrix algebra and certify both against `cov()` and `cor()`.

R code

| Stage 6 — the trap (* — run if the clock allows; cut this one first)

Q19. The notes claim `corrplot()` accepts `cov()` output and quietly draws a wrong picture: predict what actually happens.

guess

Q20. Run `corrplot(S)`, read the result aloud, and name the habit that kills this whole class of error.

R code

5. Conclusion — write it in your own words

For two variables, $|\mathbf{R}| = \text{_____}$, so it runs from 1 (nothing shared) down to 0 when

On the real data $|\mathbf{R}| = 0.0012$. Removing `life_exp` multiplies it by $\approx \text{_____}$; the factor for `health_exp_pct` is only $\approx \text{_____}$. So the redundancy is owned by

Under a change of units, $\mathbf{S}$, $\text{tr}(\mathbf{S})$ and $|\mathbf{S}|$ change by

while $\mathbf{R}$ does not — so any conclusion that must survive a units choice is read from _____.

6. Next lecture

Sunday, Aug 30 — Lecture 3, the first R lab. Bring your laptop: R, RStudio and Quarto from zero, ending in your first reproducible report. And the course's opening question is still open: a country is eight numbers, and nothing yet measures *how far* one sits from the centre of the cloud in a way that respects what $\mathbf{S}$ knows — that is Lecture 5. Qatar is in this data set, and it is one of the answers.